

BLUME CATALOG HEALTH REVIEW

Pricing, Inventory & Portfolio Risk Assessment

Business Analytics Capstone Project

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Data source: live Shopify storefront snapshot of meetblume.com (34 SKUs), captured June 2026 via Apify.

1. Executive Summary

Blume is a small, internationally-shipping Canadian skincare and wellness brand, sold direct-to-consumer and through retail partners such as Ulta Beauty. This report analyzes a live snapshot of Blume's 34-product online catalog to assess portfolio health across three dimensions: pricing strategy, inventory availability, and listing data quality.

The analysis combined SQL-based diagnostics, Python visualization, an Excel scenario model, and an interactive Power BI dashboard to identify where the catalog is underperforming and to quantify the financial impact of three prioritized recommendations.

Headline Findings

- **Over half the catalog is “at risk.”** 18 of 34 SKUs (53%) fall into a Watch, High Risk, or Critical category — driven by deep discounting, stockouts, or both.
- **The bundle product line is both the most discounted and the most stock-constrained segment.** 84.6% of “Blume”-line (bundle) products are on sale, versus 23.8% for standalone “blume-box” items — and bundles carry a 27.3% stockout rate versus 5.6% for standalone skincare.
- **Two products are discounted 83.3% — the steepest markdowns in the catalog.** Meltdown Acne Oil Treatment and Meltdown Gel Cream with Ceramides share this discount level despite very different price points, suggesting a deliberate clearance tier rather than a one-off pricing error.
- **Five products (15% of the catalog) exist as duplicate listings,** fragmenting sales and inventory data across multiple SKU records for the same item.

Estimated Financial Impact

Under a baseline assumption of 30 units sold per SKU per month (sensitivity-tested between 10 and 50 units/month):

- Aligning the Blume bundle line's discount rate to the blume-box benchmark could recover approximately **\$73,800 per year** (range: \$24,600 – \$123,000 depending on sales volume).
- Resolving stockouts on the 3 affected bundles represents approximately **\$60,900 per year** in exposure (range: \$20,300 – \$101,500).

Top 3 Recommendations

- Consolidate the 5 duplicated SKU listings into single product records (High Priority).
- Review the Blume bundle line's pricing and inventory strategy — cap discounting and add bundle-level stock monitoring (High Priority).
- Investigate the 83.3% discount tier to confirm whether it reflects intentional clearance or a reference-price data issue (Medium Priority).

2. Problem Statement & Objectives

Business Problem

Blume's operations and merchandising teams have observed inconsistent stock availability and a heavy reliance on discounting across the product catalog. This raises three open questions for the business: which products are at risk of stockout, which categories or product lines are over-discounted relative to others, and whether the catalog contains redundant or duplicated listings that could be distorting inventory and sales reporting.

Objectives

This analysis was scoped to answer four questions using the live catalog snapshot:

- Which products and categories carry the steepest discounts, and how do these compare across the catalog?
- Are there duplicate or redundant SKU listings that fragment sales and inventory data?
- Which product categories and vendor lines have the highest stockout rates, and is there a relationship between bundles and their component products?
- What is the estimated financial impact of addressing these issues, and what should be prioritized first?

Scope & Data Source

The dataset is a live snapshot of Blume's Shopify storefront (meetblume.com), retrieved via an Apify scraping actor in June 2026. It contains 34 SKUs spanning skincare products, lip balms, bundles/kits, PR kits, and one event listing. Each record includes the product title, vendor grouping, product type, pricing (current and "compare-at"), sale status, stock availability, and variant count.

This is a point-in-time snapshot, not a transactional history — a constraint that shaped the methodology described in the next section.

3. Data & Methodology

Tools Used

Tool	Role in this project
SQL Server	Data storage, cleaning, and 7 diagnostic queries (pricing, discounts, duplicates, stockouts, price spread, vendor comparison, risk classification)
Python (pandas, matplotlib, seaborn)	Exploratory data analysis, 4 visualizations, and a bundle-dependency investigation
Excel	Formula-driven “what-if” scenario model for revenue impact estimation
Power BI	4-page interactive dashboard with KPI cards, category breakdowns, and a live revenue-recovery slider

Data Cleaning Decisions

- Boolean fields (on_sale, fully_out_of_stock) were stored as 0/1 integers; this was confirmed during Power BI integration and handled accordingly in all DAX measures and Python comparisons.
- $\text{discount_pct} = (\text{compare_at_price} - \text{price_min}) / \text{compare_at_price} \times 100$, calculated only for products with a compare-at price. Products not on sale have no compare-at price and therefore a blank (NULL) discount_pct — these are correctly excluded from average-discount calculations rather than treated as zero.
- Five products were identified with duplicate title entries (different listings for the same product name); these were retained as separate rows for analysis but flagged explicitly as a data-quality finding (Section 4.2).

Risk Classification Logic

A single derived field, risk_status, was created in SQL (Query 8) to synthesize the stockout and discount findings into one classification used throughout the Python visualizations and the Power BI dashboard:

Condition	Risk Status
fully_out_of_stock = 1	Critical - Out of Stock
on_sale = 1 AND discount_pct >= 50	High Risk - Deep Discount
on_sale = 1 (discount < 50%)	Watch - On Sale
Otherwise	Healthy

Methodology Limitation: Snapshot vs. Transactional Data

The original analysis plan included time-series queries (year-over-year trends, inventory turnover ratios) and a supplier/SKU clustering model. Because the dataset is a single live snapshot with no historical order or stock-movement data, these were adapted to cross-sectional analyses better suited to the available data: a price-spread comparison (Query 6), a vendor-line comparison (Query 7), and a price-vs-discount scatter plot in place of clustering.

This adaptation is noted explicitly because recognizing data limitations and adjusting the analysis plan accordingly is itself part of the deliverable.

Similarly, the bundle-dependency analysis (Section 4.4) could not rely on product description text — which was not present in the scraped dataset — and instead used a manually researched component map, reviewed directly from Blume's product pages.

4. Key Findings

4.1 Portfolio Risk Overview

Classifying all 34 SKUs using the risk_status logic above shows that 53% of the catalog (18 products) requires some form of attention:

Risk Tier	Count	% of Catalog
Healthy	16	47.1%
Watch - On Sale	7	20.6%
High Risk - Deep Discount	6	17.6%
Critical - Out of Stock	5	14.7%

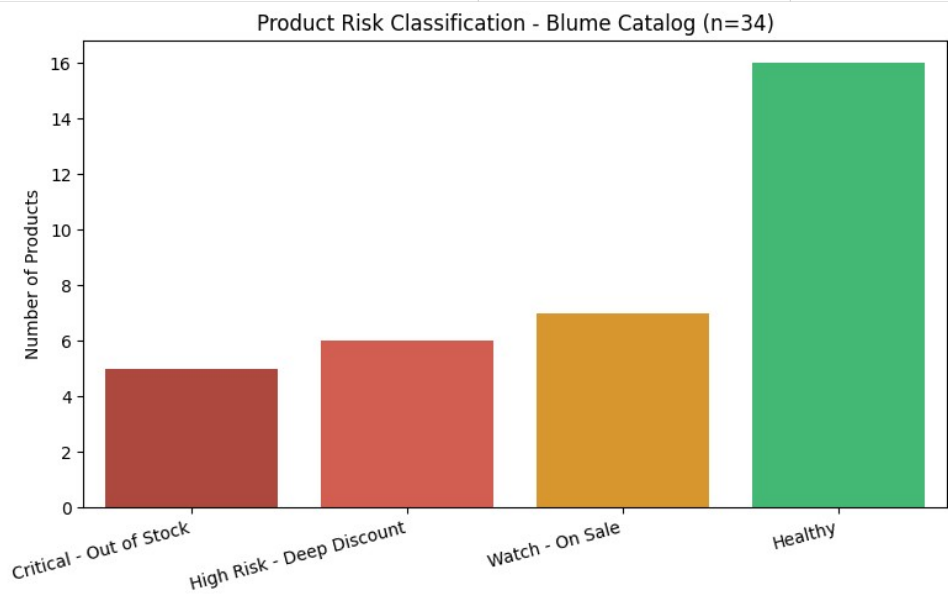


Figure 1: Product risk classification across the Blume catalog (n=34).

4.2 SKU Duplication

A query grouping products by title and counting listings (Query 3) identified five product names that appear as multiple separate listings:

Product	Listings	Price Range
Meltdown Acne Oil Treatment	2	\$52.00 – \$179.00
Meltdown Daily Balancing Gel Cleanser	2	\$32.00 – \$159.36
Meltdown Gel Cream with Ceramides	2	\$28.00 – \$110.00
Sunburst Mineral SPF 50 Sunscreen	2	\$40.00 – \$96.00
Superbalm Tripeptide-1 Vanilla Lip Balm	2	\$20.00 – \$20.00

Each duplicate represents the same physical product tracked under two separate SKU records — typically a discounted multi-variant listing alongside a single-unit, full-price listing. This fragments sales history and inventory counts for what should be one product, which can distort “best seller” rankings and complicate demand forecasting.

4.3 Pricing & Discount Analysis

The discount percentage of on-sale products varies considerably by category. Skin Care shows the widest spread — from 10% up to 83.3% — while Bundle discounts cluster more tightly between roughly 10% and 50%.

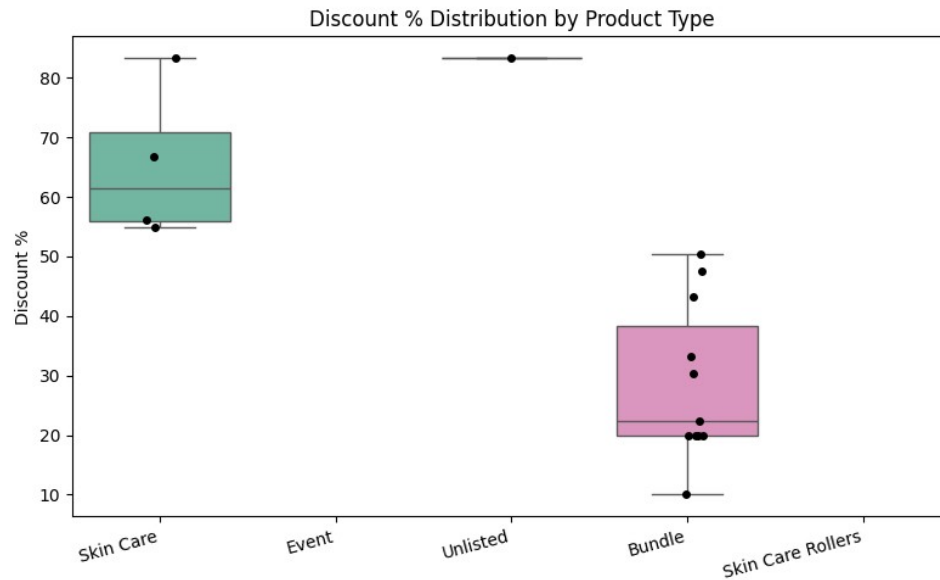


Figure 2: Distribution of discount percentages by product type, with individual products overlaid.

Plotting price against discount percentage, colored by risk tier, highlights the two products discounted at exactly 83.3% — Meltdown Acne Oil Treatment (\$52) and Meltdown Gel Cream with Ceramides (\$28). Despite their very different price points, both share the same discount rate, which points toward a deliberate clearance pricing tier applied across the catalog rather than an isolated pricing error on a single product.

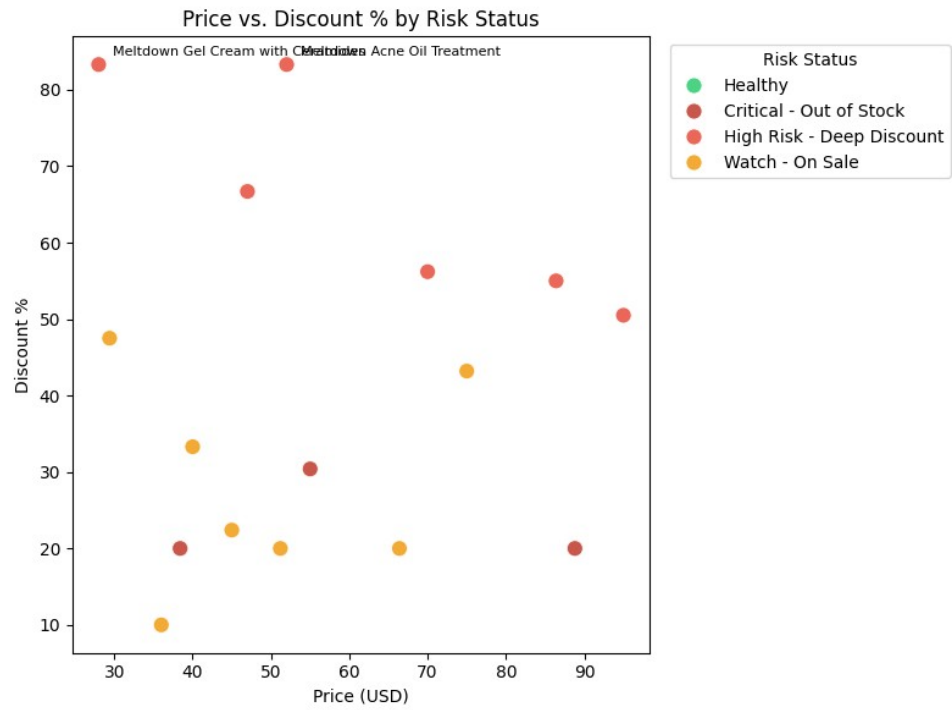


Figure 3: Price vs. discount percentage by risk status, with the two 83.3% outliers labeled.

4.4 Vendor-Line Divergence

Blume's catalog is organized under two vendor groupings: "Blume" (primarily bundles and kits) and "blume-box" (primarily standalone skincare items). Comparing these two groups reveals the most significant pattern in the dataset:

Metric	Blume (Bundles)	blume-box (Standalone)
Total SKUs	13	21
On Sale	11 (84.6%)	5 (23.8%)
Out of Stock	4	1
Average Price	\$88.86	\$58.92
Avg. Discount (on-sale items)	28.8%	68.9%

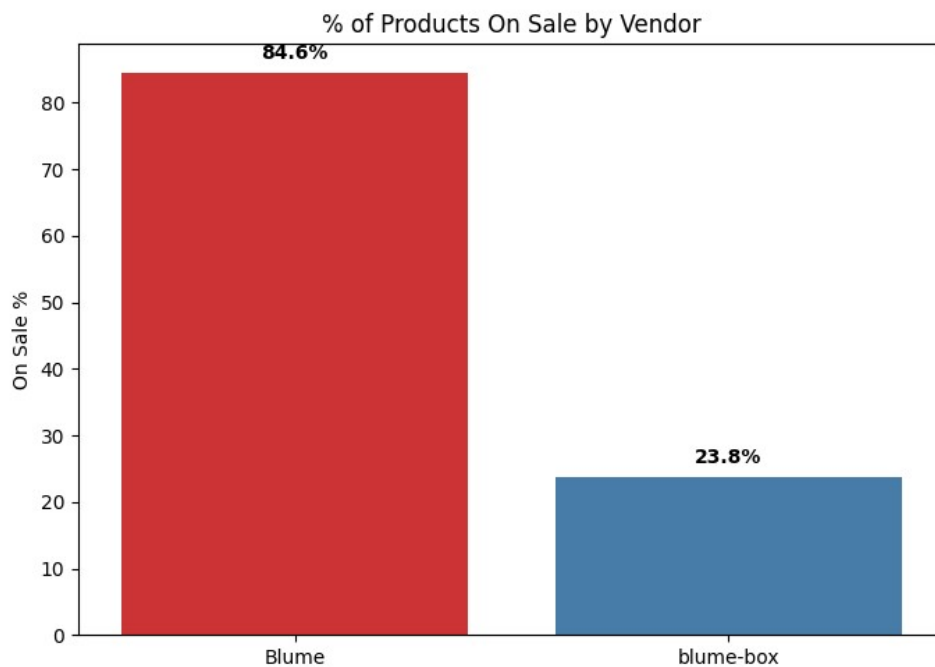


Figure 4: Percentage of products on sale, by vendor line.

The Blume bundle line is on sale at nearly 4 times the rate of the blume-box standalone line, and accounts for 4 of the catalog's 5 fully out-of-stock products despite having fewer total SKUs (13 vs. 21). This combination — high discount frequency and high stockout rate concentrated in one product line — is the strongest single signal in the dataset and the basis for Recommendation 2 (Section 6).

4.5 Stockout Risk by Category

Grouping products by category and calculating the share that are fully out of stock (Query 4) confirms that stockouts are concentrated in Bundles:

Product Type	Total Products	Out of Stock	OOS Rate
Event	1	1	100.0%
Bundle	11	3	27.3%
Skin Care	18	1	5.6%
Skin Care Rollers	1	0	0.0%
Unlisted	3	0	0.0%

Excluding the single “Event” listing (a one-off, not a recurring product category), Bundles are out of stock at roughly 5 times the rate of Skin Care products (27.3% vs. 5.6%).

4.6 Bundle Component Dependency

To understand whether bundle stockouts are caused by a shared component running out, a manual dependency map was built for the 5 bundles by reviewing their “what's included” listings on Blume's site, then cross-referenced against which individual products are marked fully out of stock.

Result: none of the mapped components for Glass Skin Kit, Hydration Heroes Kit, or Fresh Face Edit — the three out-of-stock bundles — are themselves out of stock as standalone products. This is itself a meaningful finding: these bundles appear to be out of stock independently of their components, at the bundle-SKU level. Two explanations are plausible: (1) bundle-level inventory is tracked separately from component inventory, meaning a bundle can show as sold out due to a packaging or assembly bottleneck even when every ingredient is available individually, or (2) these specific bundles have been paused or discontinued while their components remain on sale separately. Either way, the finding redirects the investigation toward bundle-level inventory management rather than raw-material shortages.

5. Interactive Dashboard

The findings above are consolidated into a 4-page interactive Power BI dashboard, allowing stakeholders to explore the catalog by risk tier, category, and vendor line, and to model revenue-recovery scenarios using an adjustable assumption slider.

Page 1 — Executive Summary

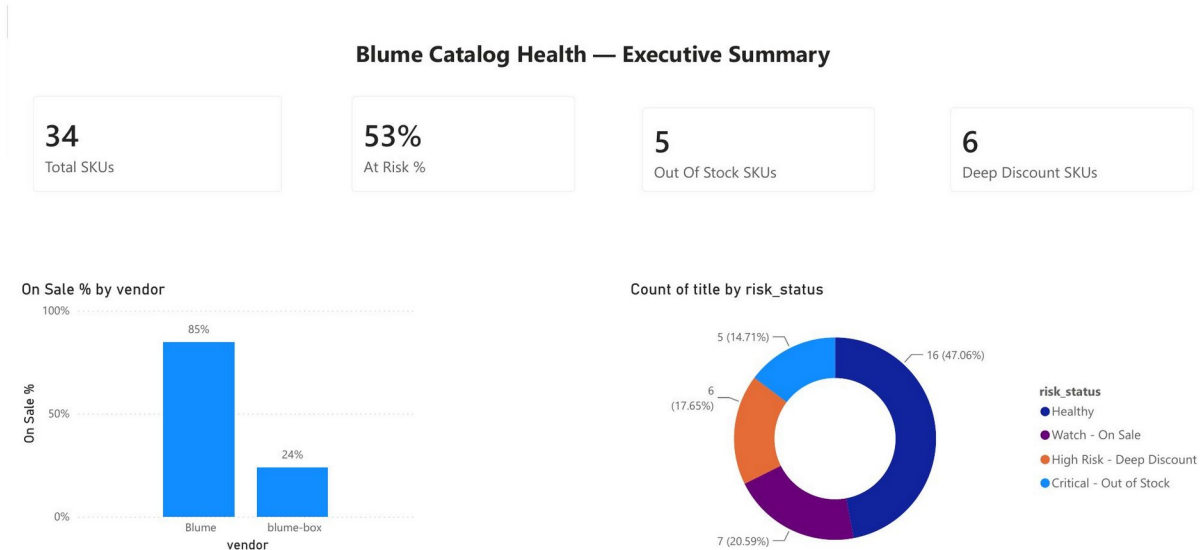


Figure 5: Dashboard Page 1 — portfolio KPIs, vendor on-sale comparison, and risk distribution.

Page 2 — Pricing & Discount Analysis



Figure 6: Dashboard Page 2 — average discount by category and a severity-colored product table.

Page 3 — Stockout & Risk Overview

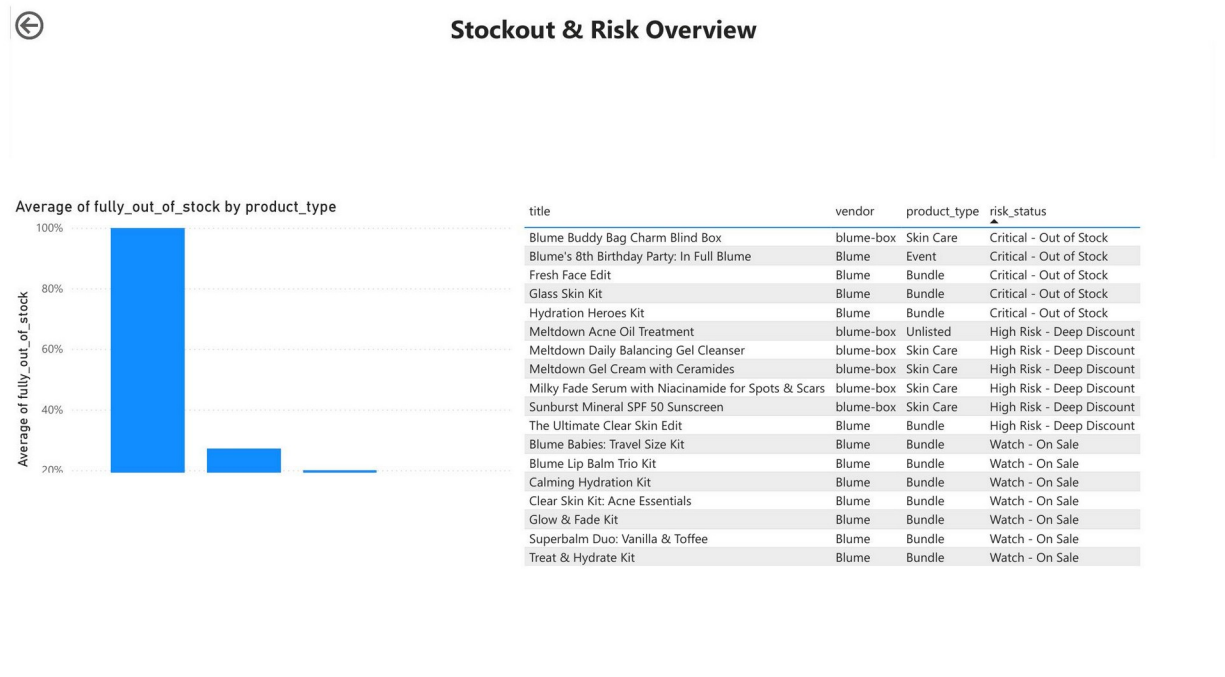
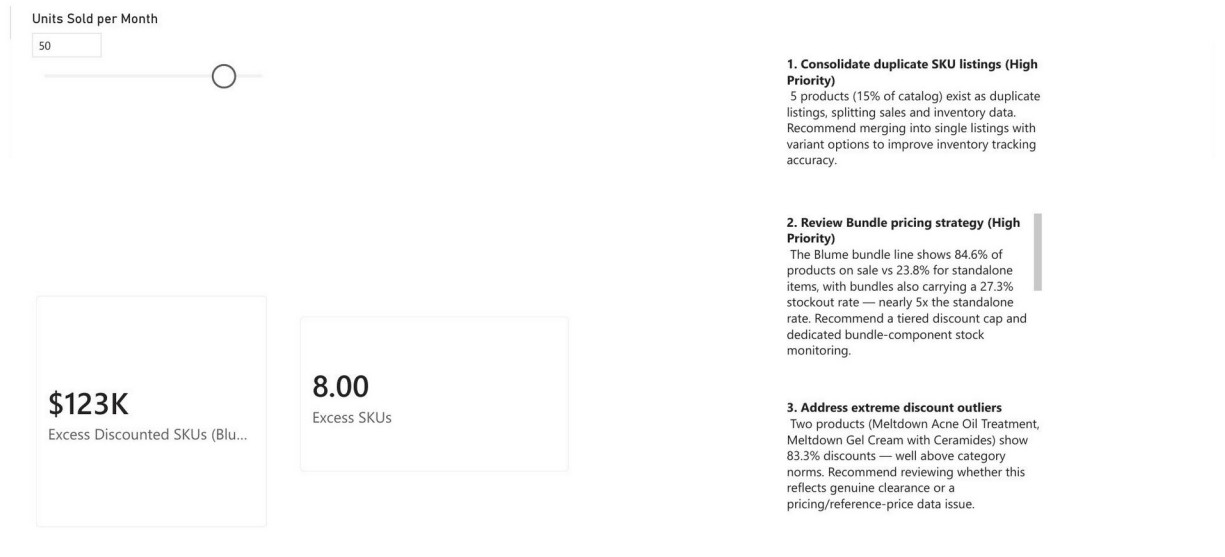


Figure 7: Dashboard Page 3 — stockout rate by category and an at-risk product list.

Page 4 — Recommendations & Revenue Recovery Scenario



6. Business Impact & Financial Modeling

To translate the findings above into financial terms, a formula-driven Excel model was built using two scenarios derived directly from the SQL and Power BI findings. Both scenarios depend on a single adjustable assumption — estimated units sold per SKU per month — since the dataset is a stock snapshot with no transaction history. A baseline of 30 units/month is used, with a sensitivity range of 10–50 units/month.

Blume — Markdown & Stockout Exposure Model

Source: live Shopify catalog snapshot, 34 SKUs (see Data tab)

ASSUMPTIONS (editable inputs — blue cells)		
Estimated avg units sold per SKU per month		30
Inventory holding cost rate (annual, % of stock value)		22.0%
Benchmark on-sale rate (blume-box vendor line, Query 7)		23.8%
CATALOG SNAPSHOT (from Data tab — Query-derived)		
Total SKUs in catalog		34
Total catalog value (sum of price_min)	\$2,392.60	
Average price per SKU	\$70.37	
SKUs on sale (all vendors)		16
SKUs fully out of stock		5
VENDOR-LINE COMPARISON (Query 7 finding)		
	Blume (bundles)	blume-box
Total SKUs	13	21
SKUs on sale	11	5
On-sale rate	84.6%	23.8%
Avg price	\$88.86	\$58.92
Avg discount % on on-sale SKUs	28.8%	68.9%
SCENARIO — IF BLUME LINE'S ON-SALE RATE MATCHED THE BENCHMARK		
Excess SKUs currently on sale vs benchmark		8
Avg discount amount per excess SKU (\$)	\$25.63	
Monthly revenue recoverable (excess SKUs × discount \$ × units sold)	\$6,151.80	
Annualized revenue recoverable		\$73,821.64
STOCKOUT EXPOSURE (Query 4 finding)		
Bundle SKUs out of stock (Query 4: 3 of 11)		3
Avg bundle price	\$56.38	
Estimated monthly revenue lost to OOS bundles	\$5,074.36	
Annualized revenue lost to OOS bundles		\$60,892.36

Note: 'units sold per month' (B5) is an assumption since the dataset is a point-in-time catalog snapshot with no transaction history.

Figure 9: Excel “Markdown & Stockout Exposure Model” — all values shown (except the three blue assumption cells) are live formulas linked to the Data tab.

Scenario 1: Bundle Discount-Rate Alignment

If the Blume bundle line's on-sale rate (84.6%) were brought in line with the blume-box benchmark (23.8%), 8 fewer bundle SKUs would be discounted. Multiplying this “excess discounted SKU” count by the average discount amount and the assumed monthly sales volume yields:

Units Sold / Month	Monthly Recoverable	Annual Recoverable
10 (conservative)	\$512	\$24,607
30 (baseline)	\$6,152	\$73,822
50 (optimistic)	\$10,253	\$123,036

Scenario 2: Bundle Stockout Resolution

The 3 currently out-of-stock bundles have an average price of \$56.38. Restocking these bundles — informed by the finding in Section 4.6 that the stockout is bundle-level, not component-level — represents the following estimated revenue recovery:

Units Sold / Month	Monthly Recoverable	Annual Recoverable
10 (conservative)	\$1,691	\$20,297
30 (baseline)	\$5,074	\$60,892
50 (optimistic)	\$8,457	\$101,487

Combined Opportunity

At the baseline assumption (30 units/SKU/month), the two scenarios together represent approximately \$134,700 per year in estimated revenue impact. These figures are illustrative — they depend on the units-sold assumption and should be validated against Blume's actual sales data before being used for planning. Their value lies in showing relative scale and in providing a reusable model: as real sales data becomes available, the single assumption cell can be replaced with actual figures and every downstream number recalculates automatically.

7. Recommendations

Recommendation 1: Consolidate Duplicate SKU Listings

Priority: High | Timeline: 2–4 weeks | Owner: Catalog / Merchandising Team

Merge the 5 duplicated product listings identified in Section 4.2 into single product records using Shopify's variant structure, so that each physical product has one SKU record with multiple price/size variants rather than two competing listings.

Expected Impact:

- More accurate sales and inventory reporting per product.
- Improved demand forecasting, since sales history is no longer split across listings.
- Reduced customer confusion and improved SEO, since duplicate listings currently compete for the same search traffic.

Recommendation 2: Review Bundle Pricing & Inventory Strategy

Priority: High | Timeline: 4–8 weeks | Owner: Pricing & Operations Teams

Address the divergence identified in Sections 4.4–4.6: the Blume bundle line is discounted at nearly 4 times the rate of standalone products (84.6% vs. 23.8%) and carries a stockout rate roughly 5 times higher (27.3% vs. 5.6%). Two concrete actions are recommended:

- Introduce a tiered discount cap for bundles, with a target on-sale rate closer to the 23.8% standalone benchmark, reviewed quarterly.
- Implement bundle-level stock monitoring as a distinct inventory category, separate from component-level stock — since Section 4.6 found that bundle stockouts are not explained by component shortages.

Expected Impact:

- Estimated \$24,600 – \$123,000 per year from discount-rate alignment (Scenario 1).
- Estimated \$20,300 – \$101,500 per year from resolving bundle stockouts (Scenario 2).

Recommendation 3: Investigate the 83.3% Discount Tier

Priority: Medium | Timeline: 1–2 weeks | Owner: Pricing Team

Two products — Meltdown Acne Oil Treatment and Meltdown Gel Cream with Ceramides — share an identical 83.3% discount despite very different price points (\$52 and \$28 respectively). Section 4.3 found this pattern consistent with a deliberate clearance tier rather than coincidence. Confirm with the pricing team whether this reflects:

- Intentional clearance ahead of reformulation or discontinuation, in which case no action is needed beyond monitoring sell-through; or
- An inflated “compare-at” reference price that overstates the discount — a common retail pricing-data issue that, if present, should be corrected to maintain pricing credibility with customers.

Expected Impact:

- Low direct revenue impact, but addresses a pricing-integrity risk and informs whether the same review should be applied to other products.

Appendix: SQL Queries

All queries were written and executed in SQL Server Management Studio against the Products table loaded from the 34-SKU catalog snapshot.

Query 1 — Category Pricing Overview

```
SELECT product_type, COUNT(*) AS total_products, AVG(price_min) AS avg_price
FROM Products
GROUP BY product_type
ORDER BY avg_price DESC;
```

Query 2 — Discount Ranking

```
SELECT title, product_type, price_min, compare_at_price, discount_pct
FROM Products
WHERE on_sale = 1
ORDER BY discount_pct DESC;
```

Query 3 — Duplicate SKU Detection

```
SELECT title, COUNT(*) AS listing_count,
       MIN(price_min) AS lowest_price, MAX(price_max) AS highest_price
FROM Products
GROUP BY title
HAVING COUNT(*) > 1
ORDER BY listing_count DESC;
```

Query 4 — Stockout Rate by Category

```
SELECT product_type, COUNT(*) AS total_products,
       SUM(CASE WHEN fully_out_of_stock = 1 THEN 1 ELSE 0 END) AS oos_count,
       CAST(100.0 * SUM(CASE WHEN fully_out_of_stock = 1 THEN 1 ELSE 0 END) / COUNT(*)
AS DECIMAL(5,1)) AS oos_pct
FROM Products
GROUP BY product_type
ORDER BY oos_pct DESC;
```

Query 6 — Price Spread Analysis

```
SELECT title, product_type, price_min, price_max,
       (price_max - price_min) AS price_spread, variant_count
FROM Products
WHERE variant_count > 1
ORDER BY price_spread DESC;
```

Query 7 — Vendor-Line Comparison

```
SELECT vendor, COUNT(*) AS total_products,
       SUM(CASE WHEN on_sale = 1 THEN 1 ELSE 0 END) AS on_sale_count,
       CAST(100.0 * SUM(CASE WHEN on_sale = 1 THEN 1 ELSE 0 END) / COUNT(*) AS
DECIMAL(5,1)) AS on_sale_pct,
       SUM(CASE WHEN fully_out_of_stock = 1 THEN 1 ELSE 0 END) AS oos_count,
       CAST(AVG(price_min) AS DECIMAL(10,2)) AS avg_price
FROM Products
GROUP BY vendor
ORDER BY total_products DESC;
```

Query 8 — Risk Classification

```
SELECT title, vendor, product_type, price_min, on_sale, fully_out_of_stock,
       discount_pct,
       CASE
         WHEN fully_out_of_stock = 1 THEN 'Critical - Out of Stock'
```

```

    WHEN on_sale = 1 AND discount_pct >= 50 THEN 'High Risk - Deep Discount'
    WHEN on_sale = 1 THEN 'Watch - On Sale'
    ELSE 'Healthy'
END AS risk_status
FROM Products
ORDER BY
CASE
    WHEN fully_out_of_stock = 1 THEN 1
    WHEN on_sale = 1 AND discount_pct >= 50 THEN 2
    WHEN on_sale = 1 THEN 3
    ELSE 4
END;

```

title	vendor	product_type	price_min	discount_pct	risk_status
Fresh Face Edit	Blume	Bundle	55.00	30.4	Critical - Out of Stock
Glass Skin Kit	Blume	Bundle	88.80	20.0	Critical - Out of Stock
Hydration Heroes Kit	Blume	Bundle	38.40	20.0	Critical - Out of Stock
Blume's 8th Birthday Party: In Full Blume	Blume	Event	55.00	NULL	Critical - Out of Stock
Blume Buddy Bag Charm Blind Box	blume-box	Skin Care	11.00	NULL	Critical - Out of Stock
Meltdown Acne Oil Treatment	blume-box	Unlisted	52.00	83.3	High Risk - Deep Discount
Meltdown Gel Cream with Ceramides	blume-box	Skin Care	28.00	83.3	High Risk - Deep Discount
Milky Fade Serum with Niacinamide for Spots	blume-box	Skin Care	47.00	66.7	High Risk - Deep Discount
Sunburst Mineral SPF 50 Sunscreen	blume-box	Skin Care	70.00	56.2	High Risk - Deep Discount
Meltdown Daily Balancing Gel Cleanser	blume-box	Skin Care	86.40	55.0	High Risk - Deep Discount

Figure 10: Query 8 result set, top 10 rows ordered by risk severity (Critical → High Risk). NULL discount_pct values correctly indicate products with no compare-at price.

Note: an additional query slot ("Query 5") in the original analysis plan covered a time-series delay metric that assumed transactional order data. Since the dataset is a single snapshot, this query was not applicable and was superseded by Queries 6 and 7 (Section 3, Methodology Limitation).